

Tianshu Zhang 张天舒

Water Resources Engineering | Hydrological & Hydraulic Modelling | GIS

ti1207zh-s@student.lu.se · zhangtt19.github.io · github.com/zhangtt19

MSc student at Lund University with experience in reservoir forecasting, river-network modelling, flood risk analysis and geospatial data processing. Experience spans field observations, data reconstruction, model calibration and validation, deployment and technical reporting.

EDUCATION

Lund University, LTH | MSc, Water Resources Engineering

September 2025 – June 2027 (expected) · Lund University Global Scholarship

Hohai University | Master's degree, Hydrology and Water Resources

September 2022 – June 2025 · College of Hydrology and Water Resources

Hohai University | Bachelor's degree, Dayu College (Honors College)

September 2018 – June 2022 · Led national- and university-level student innovation projects; both received excellent completion ratings.

ENGINEERING & RESEARCH EXPERIENCE

Tangma Reservoir forecasting and operation system | 2023–2024

Independent technical lead. Built a distributed Xin'anjiang model for a 39.9 km² catchment, coupling Muskingum routing with water balance. Reconstructed missing inflow using water-level–storage relationships and outflow records.

Led field reconnaissance, calibration, validation and deployment at the Changzhou hydrology authority. Completed technical presentations and project acceptance, followed by model adjustment and troubleshooting support.

Taihu Basin river-network change and flood risk | 2022–2025

Core modelling and analysis contributor. Updated topology and gate–pump operating conditions in a TBM model with 1,458 channels, 10,682 cross-sections and 477 hydraulic structures. Compared three operating strategies at 44 key sections.

Peak water-level absolute errors at three stations were 2–6 cm during the 2018 validation period. Prepared the Huxi-area technical report and integrated results from three study areas into the final report.

Field hydrology

Participated in more than 50 field experiments, including artificial rainfall simulation and field measurements. Maintained evapotranspiration-monitoring equipment and small weather stations.

TOOLS & LANGUAGES

Modelling: HEC-RAS, MIKE+, TBM, GMS/MODFLOW, SEEP2D, PHREEQC, AQTESOLV.

Data & GIS: Python, ArcGIS Pro, QGIS, MATLAB, R, SPSS.

Languages: Chinese; English (IELTS 7.0, Listening 8.0).

SELECTED COURSE PROJECTS · LUND UNIVERSITY

Ili–Balkhash land-cover and water-use analysis | Spring 2026

GIS and data analysis lead. Integrated 41 years of GLEAM evapotranspiration and 29 years of CCI land-cover data using Python, ArcGIS and ModelBuilder. Built a 13-class land-cover transition matrix; applied Theil–Sen estimation and Mann–Kendall tests to annual and seasonal changes.

River hydraulics and urban drainage | 2025–2026

Project lead. Built a 1D HEC-RAS model of a 9 km reach of the Rönne River and compared four discharge–sea-level combinations. Used MIKE+ to assess 85% impervious cover, 1.3× rainfall and varying outlet levels, comparing pipe resizing and storage-tank configurations.

Extreme floods and lake storage | Autumn 2026

Extreme-event analysis lead. Evaluated 88 rainfall and snowmelt scenarios for a 340 km² catchment using a conceptual rainfall–runoff model and shared group-calibrated parameters. Performed lake routing in Python for 8, 16 and 32 km² lake areas. For the selected events, the 16 km² scenario reduced simulated peaks by approximately 49–57%.

Outfall engineering design | Autumn 2026

Project lead. Compared 15 nozzle–tunnel designs for a 20 m³/s cooling-water outfall, balancing dilution, head loss and annualised cost. Under the course assumptions, 10 nozzles and a 20 m² main tunnel were the lowest-cost tested option. For a separate 6 m³/s marine outfall, calculated a 306-nozzle, 915 m diffuser and evaluated two density profiles.

SELECTED PUBLICATIONS & PATENT

Zhang, T., Jin, J., Qian, Y., et al. (2025). Research on Dynamic Trend Prediction Method for Flow Discharge Through Harbor Gates in Tidal Reaches. *Water*, 17(9), 1248.

陈钢, 张天舒, 俞悦, 等. (2025). 非线性马斯京根法在山区小流域数字孪生中的应用. *水资源保护*, 41(6), 12–17.

Chen, G., Yu, Y., Zhang, T., et al. (2025). A Distributed Unit Hydrograph Modeling for Flood Simulation in the Plain River Network Regions. *Journal of the American Water Resources Association*, 61(3), e70029.

俞悦, 张天舒, 方立超, 等. (2025). 感潮河段港闸流量动态变化计算方法研究. *水资源与水工程学报*, 36(5), 135–143.

Granted invention patent (co-inventor): CN119578220B (2025).

一种堰闸过流流量动态预测方法及系统